Autonomous Vision & Behaviour Understanding



Shadow squad

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What We are Building:

Finding objects in video is easy – understanding what they're doing is hard. Build a vision system that finds people or objects, tracks them (knows it's the same one even if it moves), and understands their behavior. Spot if behavior is normal or unusual, and surface meaningful events (not just "there's a person").

Pick a scenario: workplace safety, retail, factory, campus, traffic, or crowds.
Example: A person walking around a warehouse is normal. A person standing still in one spot for 10 minutes is unusual.

Scenario:

A person enters a shopping mall and walks around a clothing/electronics store. An autonomous vision system continuously observes the person's movements through CCTV cameras.

The system detects a sequence of unusual actions:

1. **Person enters the store** – The system detects and tracks the person.
2. **Person picks up an item** – Computer vision identifies the interaction with the product.
3. **Person looks around** – The system analyses body movement and surroundings.
4. **Person hides the item** – The system detects suspicious interaction between the person and the product.
5. **Person moves toward the exit** – The system continues tracking the person.
6. **No payment is detected** – If integrated with the store's checkout system, the system can check whether the item was purchased.
7. **Suspicious behaviour detected** – The system generates an alert for mall security.
8. **Security verification** – A security employee reviews the situation and decides whether intervention is necessary.

System Workflow

- 1. Video / camera input
- 2. Detect people or objects
- 3. Track each entity across video frames
- 4. Analyze movement and behavior
- 5. Classify behavior as normal or unusual
- 6. Record the entity and event time
- 7. Display a meaningful event / alert

HTML CODE (INDEX.HTML)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport"content="width=device-width, initial-scale=1.0">
<title>VisionGuard AI</title>
<script src="https://cdn. tailwindcss.com"></script>
</head>
<body class="bg-slate-950 text-white">
import {
  AlertTriangle,
  Camera,
  Shield,
  Users,
  Activity,
} from "lucide-react";

export default function App() {
  return (
    <div className="min-h-screen bg-slate-950 text-white">
      {/* Header */}
      <header className="bg-slate-900 border-b border-slate-800 p-5">
        <div className="flex justify-between items-center">
          <div>
            <h1 className="text-3xl font-bold text-cyan-400">
              VisionGuard AI
            </h1>
            <p className="text-gray-400">
              Autonomous Vision & Behaviour Understanding
```

```
            </p>
          </div>

          <div className="bg-green-600 px-4 py-2 rounded-lg">
            12 Cameras Online
          </div>
        </div>
      </header>

      <div className="grid grid-cols-12 gap-6 p-6">

        {/* Stats */}
        <div className="col-span-12 grid md:grid-cols-4 gap-4">

          <div className="bg-slate-900 p-5 rounded-xl">
            <Users size={40} className="text-cyan-400" />
            <h3 className="mt-3 text-gray-400">
              People Tracked
            </h3>
            <p className="text-3xl font-bold">18</p>
          </users>

          <div className="bg-slate-900 p-5 rounded-xl">
            <AlertTriangle size={40} className="text-red-500" />
            <h3 className="mt-3 text-gray-400">
              Suspicious Events
            </h3>
            <p className="text-3xl font-bold">3</p>
</alerttriangle>
          </div>

          <div className="bg-slate-900 p-5 rounded-xl">
            <Camera size={40} className="text-green-400" />
            <h3 className="mt-3 text-gray-400">
              Active Cameras
            </h3>
            <p className="text-3xl font-bold">12</p>
        </camera>
```

```
  <div className="relative">

      src="/warehouselassName="absolute left-12 top-20 border-4 border-green-500 w-24 h-44">
      <span className="bg-green-500 text-white px-2 py-1">
        ID:1 Walking
      </span>
    </div>

    <div className="absolute left-80 top-24 border-4 border-green-500 w-24 h-40">
      <span className="bg-green-500 text-white px-2 py-1">
        ID:2 Walking
      </span>
    </div>

    <div className="absolute right-32 bottom-28 border-4 border-red-500 w-40 h-64">
      <span className="bg-red-500 text-white px-2 py-1">
        ID:3 Suspicious
      </span>
    </div>
  </div>
</div>

{/* Alerts */}
<div className="col-span-4"></div

  <div className="bg-red-950 border border-red-500 rounded-xl p-5">

    <div className="flex items-center gap-3">
      <AlertTriangle
        size={35}
        className="text-red-500"/>

      <h2 className="text-2xl font-bold">
        THEFT ALERT
      </h2>
```

```
            <hr className="my-4 border-red-700" />

            <div className="space-y-3">
              <p>Person ID : 3</p>
              <p>Activity : Picking Item</p>
              <p>Zone : B1</p>
              <p>Risk Level : HIGH</p>
              <p>Confidence : 92%</p>
            </div>

            <button className="mt-5 w-full bg-red-600 py-3">
 Report Alert
 </button>
 </alerttriangle>
 </div>
 </html>
```

VisionGuard AI

Autonomous Vision & Behaviour Understanding

12 Cameras Online
{/* Stats */}

People Tracked

18

Suspicious Events

3

Use camera

Detection Accuracy

95%

{/* Camera Feed */}

Live Surveillance Feed

CAM-02
 ID:1 Walking
ID:2 Walking
ID:3 Suspicious
{/* Alerts */}

THEFT ALERT

Person ID : 3

Activity : Picking Item

Zone : B1

Risk Level : HIGH

Confidence : 92%

View Evidence
{/* Events */}

Recent Events

Person #1 Walking Normally
Person #2 Entered Zone B1
Theft Suspicion Detected
{/* Tracking Table */}

Live Object Tracking

ID	Behaviour	Zone	Risk	Status
1	Walking	A1	Low	Normal
2	Walking	B1	Low	Normal
3	Picking Item	B1	High	Alert

); }

Conclusion

- Autonomous Vision and Behaviour Understanding goes beyond object detection.
- The system combines detection, tracking, behaviour analysis, and event understanding.
- The first practical version can focus on one scenario and one abnormal behaviour.
- The system can later be extended to multiple behaviours and more general anomaly detection.